

Literature Status: Two Hatami Graph-Norm Questions

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Status

This is a **literature_already_answered** packet. It records that two questions in the concluding remarks of Hatami's *Graph norms and Sidorenko's conjecture*, arXiv:0806.0047, were later answered in the graph-norm literature. It is not a new proof packet.

Source Questions

Hatami defines Hölder and weakly Hölder bipartite graphs in Section 2.2. Theorem 2.9 of the source paper identifies these analytic inequalities with norm properties:

$$H \text{ is Hölder} \iff \|\cdot\|_H \text{ is always a seminorm,}$$

and

$$H \text{ is weakly Hölder} \iff \|\cdot\|_{r(H)} \text{ is always a norm.}$$

In later terminology these are the norming and weakly norming graph properties.

The concluding remarks ask, among other questions:

Is there any edge transitive bipartite graph that is not weakly Hölder?

and

Prove or disprove that hypercubes are Hölder.

Edge-Transitive Weakly Hölder Graphs

The supporting paper

Roman Král, Ta Sheng Noel Sandeep, Jan Volec, and Fan Wei, *The step Sidorenko property and non-norming edge-transitive graphs*, arXiv:1802.05007; J. Combin. Theory Ser. A 162 (2019), 34–54

explicitly states in its abstract and introduction that it shows the existence of a bipartite edge-transitive graph that is not weakly norming, answering a question of Hatami. The paper proves that most toroidal grids, including edge-transitive Cartesian products such as $C_n \square C_n$ outside the exceptional cases described there, are not weakly norming.

Thus Hatami's question has a positive existence answer: there are edge-transitive bipartite graphs that are not weakly Hölder.

Hypercubes

The supporting paper

Joonkyung Lee and Alexander Sidorenko, *On graph norms for complex-valued functions*, arXiv:2101.12145; J. London Math. Soc. 106 (2022), 1507–1538

states that Hatami proposed six open problems about graph norms and that it answers the last remaining one, namely whether the hypercubes, already known from Hatami’s work to be weakly norming, are norming. Its Theorem 1.4 states:

Q_d is not norming whenever $d > 2$.

Consequently Hatami’s question whether hypercubes are Hölder has a negative answer in every dimension $d > 2$.

Scope

This packet does not address the general Sidorenko conjecture. It also does not settle the other concluding questions from arXiv:0806.0047, including the hypergraph version of the framework, the weakly Hölder status of $K_2 \square C_6$, the graph obtained from $K_{5,5}$ by deleting a Hamiltonian cycle, or the moduli of smoothness and convexity of $r(H)$.

Search Evidence

Local cheap indexes were searched for arXiv:0806.0047, the source title, “Sidorenko”, “graph norm”, “weakly norming”, “norming”, “Holder”, and “weakly Holder”. No exact prior packet for these two Hatami questions was found at reservation time. The answer classification is based on the primary supporting arXiv papers listed above, both of which explicitly connect their results to Hatami’s 2010 graph-norm questions.

References

- [1] H. Hatami, *Graph norms and Sidorenko’s conjecture*, Israel J. Math. 175 (2010), 125–150; arXiv:0806.0047.
- [2] R. Král, T. S. N. Sandeep, J. Volec, and F. Wei, *The step Sidorenko property and non-norming edge-transitive graphs*, J. Combin. Theory Ser. A 162 (2019), 34–54; arXiv:1802.05007.
- [3] J. Lee and A. Sidorenko, *On graph norms for complex-valued functions*, J. London Math. Soc. 106 (2022), 1507–1538; arXiv:2101.12145.